UTM UNIVERSITI TEKNOLOGI MALAYSIA

SCSI 1013: Discrete Structure

CHAPTER 6

FINITE AUTOMATA

2018/2019 – SEM. 1

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Deterministic Finite Automata (DFA)

- In computer science, we study different types of computer languages, such as Basic, Pascal, and C++.
- We will discuss a type of a language that can be recognized by special types of machines.
- A deterministic finite automaton (pl. automata) is a mathematical model of a machine that accepts languages of some alphabet.

Deterministic Finite Automata (DFA)

- Deterministic Finite Automaton is a quintuple $M = \{ S, I, q_0, f_s, F\}$
  where,
  - S is a finite nonempty set of states
  - I is the input alphabet (a finite nonempty set of symbols)
  - q_0 is the initial state
  - f_s is the state transition function
  - F is the set of final states, subset of S.

Note:
Tuple: is an ordered list of elements.
Quintuple: five times as much in size; e.g., $M = \{S, I, O, q_0, f_s\}$;
sextuple: six time as much in size, e.g., $M = \{S, I, O, q_0, f_s, f_0\}$.

Example 1

- Let $M=\{ \{q_0,q_1,q_2\}, \{0,1\}, q_0, f_s, \{q_2\}\}$
  where f_s is defined as follows:

  $f_s(q_0,0)= q_1$, $f_s(q_1,1)= q_2$
  $f_s(q_0,1)= q_0$, $f_s(q_2,0)= q_0$
  $f_s(q_1,0)= q_2$, $f_s(q_2,1)= q_1$

- Note that for M:
  $S=\{q_0,q_1,q_2\}$, $I=\{0,1\}$, $F=\{q_2\}$
  q_0 is the initial state

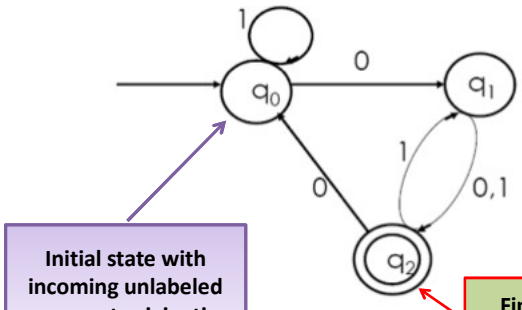
Example 1 (cont.)

- The state transition function of a DFA is often described by means of a table, called a **transition table**.

f_s	0	1
q_0	q_1	q_0
q_1	q_2	q_2
q_2	q_0	q_1

Example 1 (cont.)

- The **transition diagram** of this DFA is,



Each state represented by a small circle labeled with the state

Initial state with incoming unlabeled arrow not originating from any vertex

Final state with a double circle

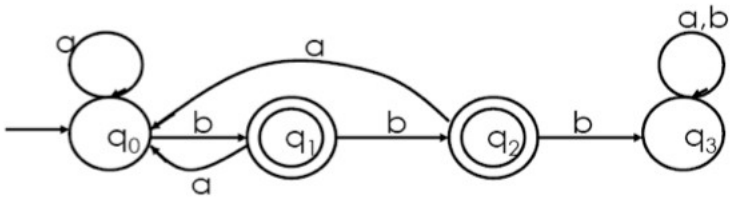
Example 2

Let $M=(\{q_0,q_1,q_2,q_3\}, \{a,b\}, q_0, f_s, \{q_1,q_2\})$ where f_s is given by the table

f_s	a	b
q_0	q_0	q_1
q_1	q_0	q_2
q_2	q_0	q_3
q_3	q_3	q_3

Example 2 (cont.)

- The transition diagram of this DFA is,



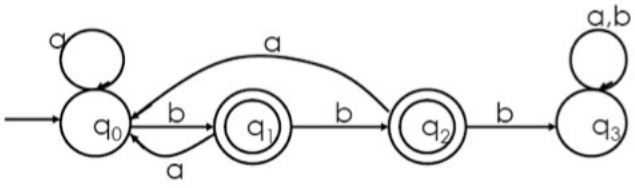
DFA – with extended transition function for M

- Let M= { S, I, q_0, f_s, F} be a DFA and w is an input string,
- w is said to be accepted by M if

$$f_s^*(q_0, w) \in F$$

- f_s^* - extended transition function for M

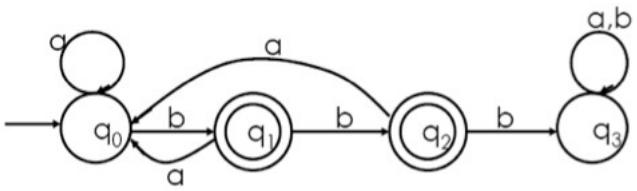
Example 3



w= abb

$q_0 \xrightarrow{a} q_0 \xrightarrow{b} q_1 \xrightarrow{b} q_2$ accepted by M

Example 4

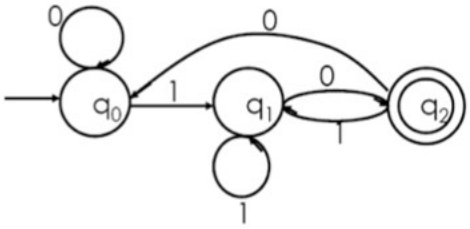


w= abba

$$q_0 \xrightarrow{a} q_0 \xrightarrow{b} q_1 \xrightarrow{b} q_2 \xrightarrow{a} q_0$$

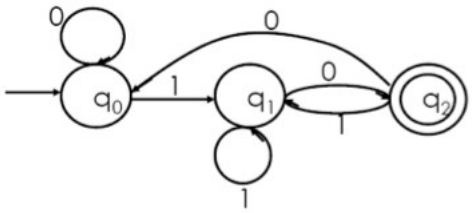
not accepted by M

Example 5



- What are the states of M?
- Write the set of input symbols.
- Which is the initial state?

Example 5 (cont.)



- Write the set of final states.
- Write the transition table for this DFA

Example 5 (cont.)

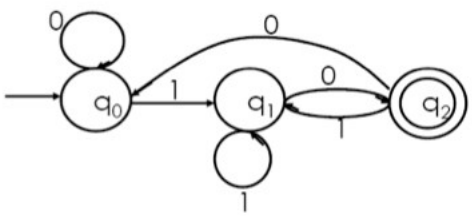
Which of the strings are accepted by M?

0111010, 00111, 111010,

0100, 1110

Example 5 - Solution

0111010

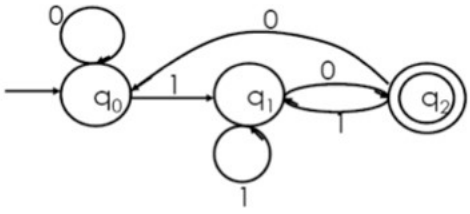


$q_0 \xrightarrow{0} q_0 \xrightarrow{1} q_1 \xrightarrow{1} q_1 \xrightarrow{1} q_1 \xrightarrow{0} q_2 \xrightarrow{1} q_1 \xrightarrow{0} q_2$

accepted by M

Example 5 – Solution (cont.)

00111

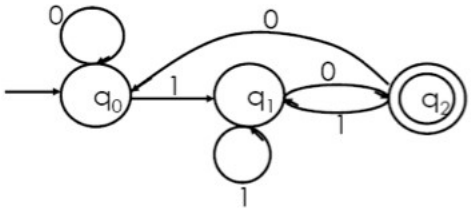


$q_0 \xrightarrow{0} q_0 \xrightarrow{0} q_0 \xrightarrow{1} q_1 \xrightarrow{1} q_1 \xrightarrow{1} q_1$

not accepted by M

Example 5 – Solution (cont.)

111010

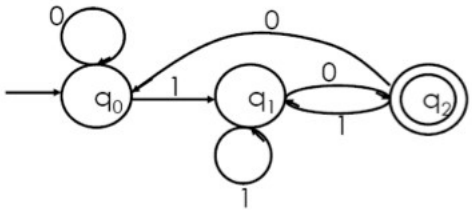


$q_0 \xrightarrow{1} q_1 \xrightarrow{1} q_1 \xrightarrow{1} q_1 \xrightarrow{0} q_2 \xrightarrow{1} q_1 \xrightarrow{0} q_2$

accepted by M

Example 5 – Solution (cont.)

0100

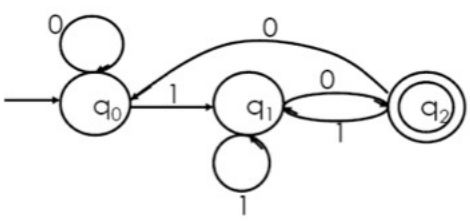


$q_0 \xrightarrow{0} q_0 \xrightarrow{1} q_1 \xrightarrow{0} q_2 \xrightarrow{0} q_0$

not accepted by M

Example 5 – Solution (cont.)

1110



$q_0 \xrightarrow{1} q_1 \xrightarrow{1} q_1 \xrightarrow{1} q_1 \xrightarrow{0} q_2$

accepted by M

Example 6

Construct a state transition diagram of a DFA that accepts on {a,b} that contain an even number of a's and an odd number of b's.

Example of accepted strings:
aab, baa, baaabba

Example 6 - Solution

4 states,

q_0	even num. of a's & even num. of b's.
q_1	even num. of a's & odd num. of b's.
q_2	odd num. of a's & odd num. of b's.
q_3	odd num. of a's & even num. of b's.

$S = \{q_0, q_1, q_2, q_3\}$

Example 6 – Solution (cont.)

set of states, $S = \{q_0, q_1, q_2, q_3\}$

set of input symbols, I= {a, b}

initial state, q_0

final state, q_1

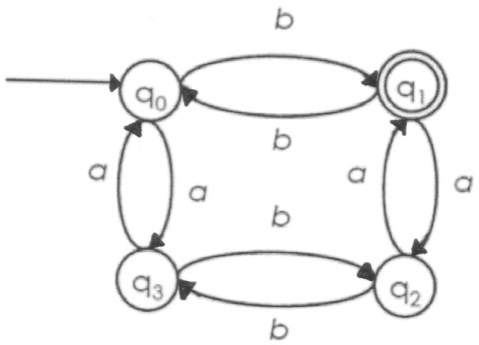
Example 6 – Solution (cont.)

State transition function

f_s	a	b
q_0	q_3	q_1
q_1	q_2	q_0
q_2	q_1	q_3
q_3	q_0	q_2

Example 6 – Solution (cont.)

State transition diagram



Exercise 1

Let M=(S, I, q_0, f_s, F) be the DFA such that S={q_0,q_1,q_2}, I={a,b}, F={q_2}, q_0 =initial state, and f_s is given by,

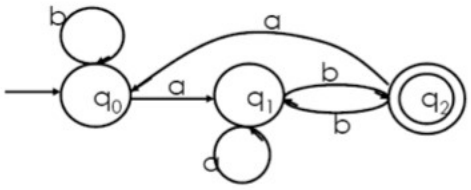
f_s	a	b
q_0	q_0	q_1
q_1	q_2	q_1
q_2	q_2	q_0

Draw the state diagram of M.

Which of the strings
abaa, bbbabb, bbbaa dan bababa
are accepted by M?

Exercise 2

The transition diagram of M is,



Construct the transition table of M.
Which of the strings
baba, baab, abab dan abaab
are accepted by M?

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Exercise 3

Construct a state transition diagram of a DFA M with the input set {0,1} such that M accepts only the string 101.

Solution:

